

Adithya Thirupathi

Software Engineer

United States

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Summary

Highly accomplished Software Developer with 3+ years of expertise specializing in Java Full-Stack development, microservices architecture, and cloud-native applications. Proven track record of designing and implementing robust, scalable enterprise solutions across financial services, healthcare, and e-commerce domains. Adept at optimizing application performance, achieving high system uptime, and delivering substantial business value (\$5M+) by supporting applications for 8M+ global users.

- Java Full-Stack Development Expert:** Proficient in Java, Spring Boot, React, and Node.js, with extensive experience in architecting and developing responsive web applications, robust RESTful APIs, and complex microservices architectures. Successfully managed 100+ production services and reduced development time by 60% through efficient Java-centric practices.
- Cloud-Native & Microservices Specialist:** Advanced skills in AWS, Azure, and containerization technologies (Docker, Kubernetes), specifically in deploying and scaling Java-based applications. Implemented highly available cloud solutions that improved performance by 80% while maintaining 99.99% availability and reducing infrastructure costs by 40%.
- Modern Development Professional:** Expertise in Agile methodologies, CI/CD pipelines (Jenkins, Git), and DevOps practices, delivering high-quality software solutions across production environments with 95% automated testing coverage (JUnit, Jest) and seamless deployment capabilities.

Skills

- Programming Languages:** Java (Core, EE), Python, JavaScript, TypeScript, C#, C++, SQL, HTML5, CSS3, PHP
- Frontend Frameworks:** React.js, Angular, Vue.js, Next.js, Bootstrap, Material-UI, Tailwind CSS, jQuery
- Backend Technologies:** Spring Boot, Node.js, Express.js, Django, Flask, .NET Core, ASP.NET MVC
- Database Systems:** MySQL, PostgreSQL, MongoDB, Oracle, SQL Server, Redis, DynamoDB, Cassandra
- Cloud Platforms:** AWS (EC2, S3, Lambda, RDS, API Gateway, CloudFormation), Azure (App Service, Functions, SQL Database), GCP (Compute Engine, Cloud Functions, Cloud SQL)
- DevOps & Tools:** Docker, Kubernetes, Jenkins, Git, GitHub, GitLab, Terraform, Ansible, CI/CD Pipelines
- Testing Frameworks:** JUnit, TestNG, Jest, Selenium, Cypress, Postman, SoapUI, Mocha, Chai
- Web Technologies:** RESTful APIs, GraphQL, WebSockets, Microservices, JSON, XML, SOAP
- Development Tools:** IntelliJ IDEA, Eclipse, Visual Studio Code, Maven, Gradle, npm, Webpack, Babel

Experience

- Software Engineer | PNC Bank, MI - USA
- OCT 2024 – PRESENT
- Enterprise Java Banking Application Development:** Architected and developed scalable web applications for \$200B+ banking operations, primarily using Java Spring Boot for backend services alongside React and Node.js for frontend integration. Implemented responsive designs and RESTful APIs that reduced page load times by 70% and improved customer satisfaction by 25%.
 - Core Banking Microservices & Java Ecosystem:** Designed and implemented 25+ critical microservices using Java, Spring Boot, and Docker for core banking systems. Established a resilient, event-driven architecture with Apache Kafka, significantly improving system reliability by 85% and reducing service coupling dependencies by 60%.
 - Real-Time Java Payment Processing:** Developed high-performance payment processing systems utilizing Java, Redis, and PostgreSQL, handling \$50M+ daily transactions. Implemented sophisticated real-time fraud detection algorithms in Java that reduced fraudulent transactions by 80% and improved processing speed by 65%.
 - Financial APIs & Integration:** Built comprehensive API gateway solutions using Node.js, Express.js, and AWS API Gateway to integrate with 15+ third-party financial services. Implemented robust security using OAuth 2.0 and JWT authentication, processing 5M+ secure transactions daily with 99.99% uptime.
 - Java Security & Regulatory Compliance:** Implemented comprehensive security frameworks using Spring Security, JWT tokens, and encryption protocols to ensure 100% compliance with banking regulations (SOX, PCI-DSS), preventing potential \$10M+ in regulatory penalties.
 - Database Optimization & Management:** Optimized complex SQL queries and database schemas across Oracle, PostgreSQL, and MongoDB systems, managing 500GB+ financial data. Implemented indexing strategies and query optimization that improved database performance by 75% and reduced query response times from 5 seconds to 200ms.

- **Automated Testing & Quality Assurance:** Established comprehensive testing frameworks using JUnit (for Java), Jest, and Selenium with 95% code coverage, implementing automated unit, integration, and end-to-end testing that reduced production bugs by 90% and improved deployment confidence.

Software Engineer | Infosys, Hyderabad - INDIA

JUN 2021 – JUN 2023

- **Multi-Industry Java-Centric Application Development:** Led full-stack development for 20+ client applications across healthcare, insurance, e-commerce, retail, and IT sectors, heavily leveraging Java and Spring Boot for robust backend solutions, alongside React and Angular for intuitive user interfaces. Delivered custom software solutions that improved business efficiency by 50% and generated \$8M+ in client value.
- **Insurance Platform Development with Java:** Built comprehensive insurance management platforms primarily using Java Spring Boot and Oracle databases, with Angular for the frontend, handling 500K+ policies annually. Implemented automated claims processing and underwriting workflows that reduced policy approval time by 55% and improved customer experience ratings by 30%.
- **E-commerce Solutions Architecture:** Developed scalable e-commerce platforms using React, Node.js, and AWS services, supporting 2M+ customers during peak shopping seasons. Implemented real-time inventory management and personalized recommendation engines that increased conversion rates by 45% and average order value by 35%.
- **Cloud Migration & Modernization (with Java focus):** Led migration of 30+ legacy applications, including significant Java applications, to AWS cloud infrastructure using containerization (Docker, Kubernetes) and serverless technologies. Implemented auto-scaling and load balancing that reduced operational costs by 45% and improved application availability to 99.9%.
- **API Development & Integration:** Designed and implemented 50+ RESTful APIs using Node.js, Express.js, and Python Django for seamless third-party integrations, establishing standardized API documentation and versioning that improved integration efficiency by 70% and reduced development time by 40%.
- **Agile Development & Project Management:** Collaborated with cross-functional teams using Agile/Scrum methodologies to deliver 15+ major software releases, implementing CI/CD pipelines with Jenkins and Git that reduced deployment time by 75% and improved code quality through automated testing.
- **Performance Optimization & Monitoring:** Implemented comprehensive application monitoring using New Relic, Grafana, and custom logging solutions, establishing performance benchmarks and optimization strategies that improved application response times by 65% and reduced server resource utilization by 35%.

Education

Master of Science in Information Systems | Central Michigan University, Mount Pleasant, MI - USA

Bachelor of Engineering in Electrical & Electronics Engineering | Mahatma Gandhi Institute of Technology, Hyderabad – INDIA

Key Projects

Banking Transaction Processing System

- Designed and implemented a high-availability transaction processing system handling \$100M+ daily transactions, built entirely with Java microservices. Achieved real-time fraud detection with 99.8% accuracy and sub-200ms response times while maintaining 99.99% uptime.

Healthcare Patient Management Platform

- Built an end-to-end insurance claims processing system using Java Spring Boot for robust backend logic and Angular for the user interface. Implemented automated claim assessment and approval workflows that reduced processing time by 70% and improved accuracy by 85%.

E-commerce Multi-Vendor Marketplace

- Architected and developed a scalable e-commerce platform using React, Node.js, and AWS, supporting 1M+ users and 10K+ vendors. Implemented real-time inventory management and an AI-powered recommendation engine.

Healthcare Patient Management Platform

- Led development of a comprehensive patient management system using React and Node.js, serving 500+ healthcare providers. Implemented HIPAA-compliant architecture with real-time patient monitoring and telemedicine capabilities.